Problem Set 4 · Optimal bundles and kinked budget lines

Microeconomics (040184) · Group 1 · WS 2026/27 · **Deadline: Wednesday 11 November 2026, 16:45**

How this problem set works

When you use an LLM, you may let it solve a problem and then go over its solution. As you read through the solution, each step makes sense, and it is easy to think that you have understood it. Whether you have really understood it is something you only find out when you try to solve a problem on your own. This problem set is an opportunity to test this for yourself.

If this is how you approach problems, which is fine, then do so here: ask your LLM to solve Problem 1, go over its solution carefully, and make sure that you understand each of its steps. Problem 1 is optional and is not part of the quiz, so use it to learn.

Then, without using the LLM, test your understanding on Problem 2. Problem 2 is a different problem, but if you have understood Problem 1 well, you should be able to solve it. If you get stuck, do not ask your LLM right away. First go back to the solution of Problem 1 and, using the solution steps there, try to figure out how to solve Problem 2 on your own. Only if you are still stuck after giving it a good amount of time, get assistance from your LLM on Problem 2 as well. In this way, you will see how far you can get on your own, and whether what you thought you understood from the LLM's solution is actually calibrated.

Problems 2 and 3 are in the quiz and are graded.

An optimization question with a single budget line is an easy question. You should reach the level where it is also very easy for you, and then you should become very comfortable with questions with a kinked budget line.

Next week we will cover problems of this difficulty with different utility functions, so it is important that you become very comfortable solving problems at the level of Problem 2 on your own.

Once you get comfortable solving problems at the level of Problem 2, the next thing you need to tackle is to convert a verbal problem into equations. In the exam, I may make the question harder by not directly providing the budget sets as equations or the utility function as an equation, but by placing them in a verbal story from which you need to extract them. So you need to become proficient at doing this as well.

In the literature, one way to think of the complexity of a problem is by the difficulty of building the equational or functional form of the problem that you need to solve. Many people fail at this even though, if they were given the functional form, they could solve the rest. So it is equally important that you also master this skill of being able to write down the functional form of the problem (the utility function and the budget sets) given a textual story. You will practice this with Problem 3.

Answers are entered in the Moodle quiz *PS4* (*Group 1*), which opens by Monday 9 November at the latest. Only the Moodle answers are graded, and each student submits individually. The deadline is **Wednesday 11 November 2026, 16:45**.

Drawing graphs will help you understand what is going on and get the general picture, but you should not rely on them to identify the optimal consumption bundle. I strongly encourage you to always draw a graph with the budget lines. Once you have found the solution, try to understand what the indifference curves must look like to give the solution you found. With practice, you should be able to picture the graph just by looking at the equations.

Problem 1 · Kinked budget lines

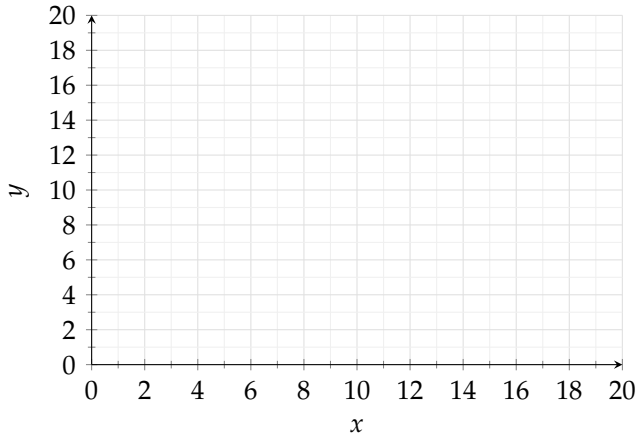
Part A · A single budget line

Notation. A **bundle** (x, y) contains x units of good x and y units of good y, and $u(x, y)$ is its **utility**. p_x and p_y are the **prices** of the two goods, and B is the **income**. The **budget set** contains all the bundles the consumer can afford: $p_x x + p_y y \leq B$. The **budget line** contains the bundles that cost exactly the income: $p_x x + p_y y = B$.

Given

A consumer has the utility function $u(x, y) = 3xy$. The prices are $p_x = p_y = €1$ and the income is $B = €10$, so the budget line is $x + y = 10$. Consider the bundles P $= (2, 8)$ and Q $= (1, 9)$.

(a) Draw the budget line in the graph below, with its end points, and mark the bundles P and Q. You will add the budget lines of Parts B and C to the same graph.



Notation. MU_x is the **marginal utility** of x: the derivative of $u(x, y)$ with respect to x, with y held fixed. MU_y is the marginal utility of y, with x held fixed. MRS is the **marginal rate of substitution**, $\text{MRS} = \frac{\text{MU}_x}{\text{MU}_y}$: the rate at which the consumer can give up y for more x and keep utility the same, as the step shrinks.

(b) At the bundle P $= (2, 8)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q $= (1, 9)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(d) The MRS is different at P and at Q. Consider these statements.

I. MU_x depends on y.
II. MU_y depends on x.
III. The prices are different at P and at Q.

Which of the statements above give the reason?

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

(e) Consider these statements about the ratio $\frac{\text{MU}_x}{p_x}$.

I. It is the "bang for the buck" on x.
II. It is the marginal utility of x per euro spent on x.
III. It is the marginal utility of x, normalized by the price of y.

Which of the statements above correctly interpret $\frac{\text{MU}_x}{p_x}$?

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

Notation. A star marks an optimal value: (x^*, y^*) denotes the optimal bundle.

(f) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part B · A change in income

Given

The utility function is still $u(x, y) = 3xy$ and both prices are still €1, but the income rises from €10 to $B =$ €18, so the budget line is now $x + y = 18$. Consider the bundles P $= (2, 16)$ and Q $= (3, 15)$.

(a) Draw the new budget line in the graph of Part A, with its end points, and mark the bundles P and Q.

(b) At the bundle P $= (2, 16)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____
(ii) The marginal utility of y: $\text{MU}_y =$ _____
(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q = $(3, 15)$, calculate the following.

(i) The marginal utility of x: MU_x = _____

(ii) The marginal utility of y: MU_y = _____

(iii) The marginal rate of substitution: MRS = _____

(d) Compare the bundle $(2, 16)$ with the bundle $(2, 8)$ from Part A. The bundle $(2, 16)$ has the same quantity of x as the bundle $(2, 8)$ and a larger quantity of y. Which of the following is correct?

- ☐ MU_x is higher at $(2, 16)$ than at $(2, 8)$, MU_y is the same at both bundles, and the MRS is higher at $(2, 16)$ than at $(2, 8)$
- ☐ MU_x is lower at $(2, 16)$ than at $(2, 8)$, MU_y is the same at both bundles, and the MRS is lower at $(2, 16)$ than at $(2, 8)$
- ☐ MU_x is the same at both bundles, MU_y is higher at $(2, 16)$ than at $(2, 8)$, and the MRS is lower at $(2, 16)$ than at $(2, 8)$
- ☐ both marginal utilities are higher at $(2, 16)$ than at $(2, 8)$, and the MRS is the same at both bundles
- ☐ both marginal utilities and the MRS are the same at both bundles

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part C · A change in price

Given

The utility function is still $u(x, y) = 3xy$ and the income is still €18, but the price of x rises from €1 to p_x = €3, so the budget line is now $3x + y = 18$. Consider the bundles P = $(1, 15)$ and Q = $(2, 12)$.

(a) Draw the new budget line in the graph of Part A, with its end points, and mark the bundles P and Q.

(b) At the bundle P = $(1, 15)$, calculate the following.

(i) The marginal utility of x: MU_x = _____

(ii) The marginal utility of y: MU_y = _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q = $(2, 12)$, calculate the following.

(i) The marginal utility of x: MU_x = _____

(ii) The marginal utility of y: MU_y = _____

(iii) The marginal rate of substitution: MRS = _____

(d) The price of x is €1 in Part B and €3 in this part. Take one bundle and keep it fixed. Which of the following has a different value at this bundle when the price of x is €3 than when it is €1?

- ☐ MU_x and the MRS
- ☐ MU_x, MU_y and the MRS
- ☐ only MU_x
- ☐ only $\frac{\text{MU}_x}{p_x}$
- ☐ only the MRS

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

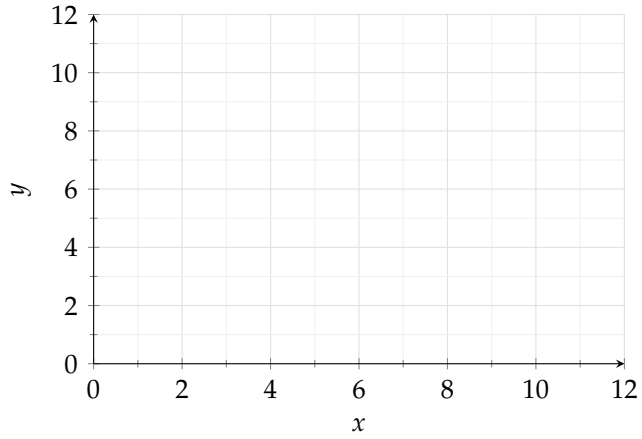
Part D · A kinked budget line

Notation. A **kink** is a point where a budget line changes slope. A **segment** is a straight part of the budget line. A **composite budget line** has one part from one budget line and one part from another budget line.

Given

The utility function is still $u(x, y) = 3xy$. The income is €10 and the price of y is €1. The first 4 units of x cost €1 each, and each further unit costs €3. The budget line therefore has two segments: $x + y = 10$ for $x \leq 4$, and $3x + y = 18$ for $x \geq 4$.

(a) Graph the composite budget line below, with its end points and its kink.



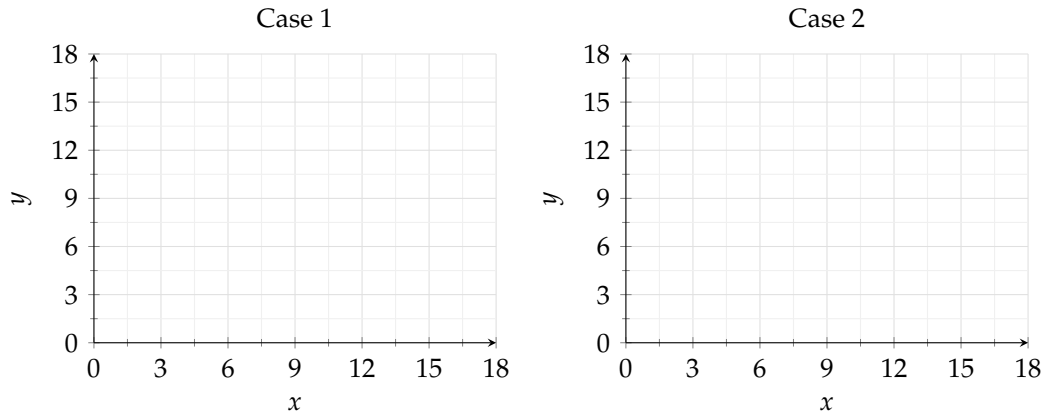
(b) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part E · Two further cases

Given

The utility function is still $u(x, y) = 3xy$ and the price of y is €1. Here are two further kinked budget lines.
Case 1. The income is €10. The first 7 units of x cost €1 each, and each further unit costs €3. The budget line is therefore $x + y = 10$ for $x \leq 7$, and $3x + y = 24$ for $x \geq 7$.
Case 2. The income is €16. The first unit of x costs €1, and each further unit costs €3. The budget line is therefore $x + y = 16$ for $x \leq 1$, and $3x + y = 18$ for $x \geq 1$.

(a) Graph the composite budget line of each case below, with its end points and its kink.



(b) Case 1. What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

(c) Case 2. What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Before Problem 2: derivatives of powers and roots

The power rule. For a power α of a variable, differentiation multiplies by α and lowers the power by 1.

$$\frac{d}{dx}\, x^{\alpha} = \alpha\, x^{\alpha-1}$$

Sums. The derivative of a sum is the sum of the derivatives. A constant factor is retained.

Negative powers. A negative power denotes a reciprocal: $x^{-\frac{1}{2}} = \dfrac{1}{\sqrt{x}}$.

A square root is a power. $\sqrt{x} = x^{\frac{1}{2}}$. So its derivative is $\frac{1}{2}\, x^{-\frac{1}{2}} = \dfrac{1}{2\sqrt{x}}$.

A constant inside the root. $\sqrt{9x} = 3\sqrt{x}$, so its derivative is $3 \cdot \dfrac{1}{2\sqrt{x}} = \dfrac{3}{2\sqrt{x}}$.

Two goods. When differentiating with respect to x, y is treated as a constant.

$$10\sqrt{xy} = 10\sqrt{y} \cdot \sqrt{x}, \quad \text{so}$$

$$\mathrm{MU}_x = 10\sqrt{y} \cdot \frac{1}{2\sqrt{x}} = 5\sqrt{\frac{y}{x}}$$

Similarly, $\mathrm{MU}_y = 5\sqrt{\dfrac{x}{y}}$. Problem 2 uses the coefficient 12 instead of 10.

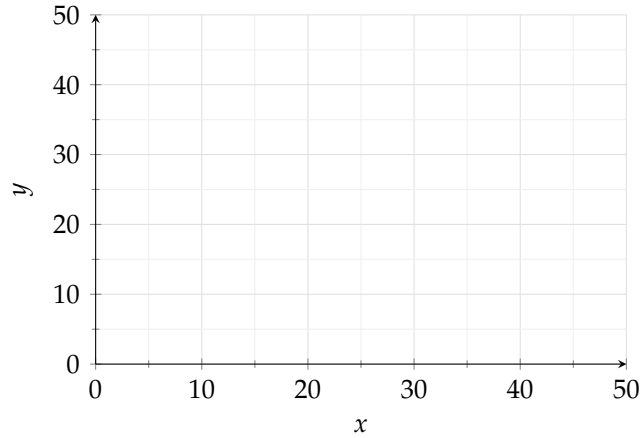
Problem 2 · Kinked budget lines and a square root

Part A · A single budget line

Given

A consumer has the utility function $u(x, y) = 12\sqrt{xy}$. The prices are $p_x = p_y = €1$ and the income is $B = €24$, so the budget line is $x + y = 24$. Consider the affordable bundles P $= (4, 16)$ and Q $= (2, 18)$.

(a) Draw the budget line in the graph below, with its end points, and mark the bundles P and Q. You will add the budget lines of Parts B and C to the same graph.



(b) At the bundle P $= (4, 16)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q $= (2, 18)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(d) The MRS is different at P and at Q. Consider these statements.

I. The ratio $\frac{y}{x}$ is different at P and at Q.

II. The coefficient 12 in the utility function is different at P and at Q.

III. Each marginal utility depends on both x and y.

Which of the statements above give the reason?

☐ only I

☐ only II

☐ I and II

☐ I and III

☐ I, II and III

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part B · A change in income

Given

The utility function is still $u(x, y) = 12\sqrt{xy}$ and both prices are still €1, but the income rises from €24 to $B =$ €48, so the budget line is now $x + y = 48$. Consider the affordable bundles P $= (8, 32)$ and Q $= (4, 36)$.

(a) Draw the new budget line in the graph of Part A, with its end points, and mark the bundles P and Q.

(b) At the bundle P $= (8, 32)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q $= (4, 36)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(d) The bundle $(4, 16)$ from Part A and the bundle $(8, 32)$ have the same MRS. What is the reason?

- ☐ the ratio $\frac{y}{x}$ is the same at both bundles
- ☐ both bundles lie on a budget line
- ☐ the income is higher in Part B than in Part A
- ☐ the prices are the same in Part A and in Part B
- ☐ the utility is the same at both bundles

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part C · A change in price

Given

The utility function is still $u(x, y) = 12\sqrt{xy}$ and the income is still €48, but the price of x rises from €1 to $p_x =$ €4, so the budget line is now $4x + y = 48$. Consider the affordable bundles P $= (1, 36)$ and Q $= (8, 8)$.

(a) Draw the new budget line in the graph of Part A, with its end points, and mark the bundles P and Q.

(b) At the bundle P $= (1, 36)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q $= (8, 8)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(d) The price of x is €1 in Part B and €4 in this part. Take one bundle and keep it fixed. Which of the following has a different value at this bundle when the price of x is €4 than when it is €1?

- ☐ MU_x and the MRS
- ☐ MU_x, MU_y and the MRS
- ☐ only MU_x
- ☐ only $\frac{\text{MU}_x}{p_x}$
- ☐ only the MRS

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

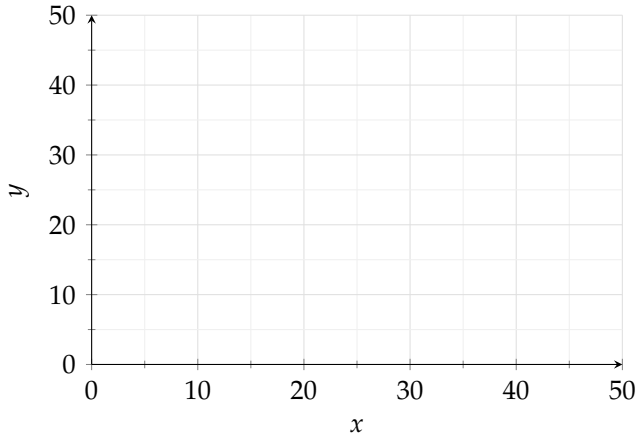
Part D · Two budget lines available

Notation. The **outer boundary** of two budget lines contains, at each x, the bundle with the highest y on either line. It is the composite budget line of the two.

Given

The utility function is still $u(x, y) = 12\sqrt{xy}$. Now the consumer can choose a bundle on either of two budget lines: $4x + y = 48$ from Part C or $x + y = 24$ from Part A. The two lines intersect, and the kink of the outer boundary, the point where they cross, lies closer to the origin than the two ends of the boundary. To the left of the kink, $4x + y = 48$ is the outer line; to the right of the kink, $x + y = 24$ is the outer line. The outer boundary is therefore $4x + y = 48$ for $x \leq 8$, and $x + y = 24$ for $x \geq 8$.

(a) Graph the composite budget line below, with its end points and its kink.



(b) Which of the following gives the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income?

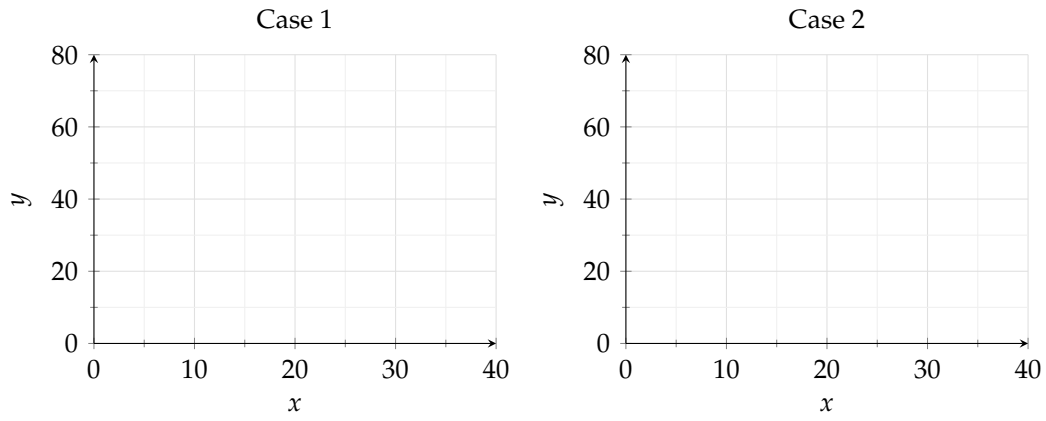
- ☐ only $(8, 16)$
- ☐ both $(4, 32)$ and $(16, 8)$
- ☐ only $(12, 12)$
- ☐ both $(6, 24)$ and $(12, 12)$
- ☐ only $(6, 24)$

Part E · Two further cases

Given

The utility function is still $u(x, y) = 12\sqrt{xy}$. In each case, the consumer can again choose a bundle on either of two budget lines.
Case 1. The two budget lines are $4x + y = 48$ and $x + y = 36$. The outer boundary is $4x + y = 48$ for $x \leq 4$, and $x + y = 36$ for $x \geq 4$.
Case 2. The two budget lines are $4x + y = 72$ and $x + y = 24$. The outer boundary is $4x + y = 72$ for $x \leq 16$, and $x + y = 24$ for $x \geq 16$.

(a) Graph the composite budget line of each case below, with its end points and its kink.



(b) Case 1. Determine the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income.

(c) Case 2. Determine the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income.

Problem 3 · From words to equations

Notation. $\min\{a, b\}$ is the smaller of the two numbers a and b.

Part A · Preferences in words

Each story below describes a consumer's preferences in words.

(a) One pack of pasta makes one meal, and the consumer is indifferent between two brands of pasta. Let x denote the number of packs of the first brand and y the number of packs of the second brand. The consumer cares only about the number of meals. Which utility function represents these preferences?

- ☐ $u = 2x + y$
- ☐ $u = \min\{x, y\}$
- ☐ $u = x + 4y$
- ☐ $u = x + y$
- ☐ $u = xy$

(b) A small pack contains 2 eggs and a large pack contains 3 eggs. The consumer cares only about the total number of eggs. Let x denote the number of small packs and y the number of large packs. Which utility function represents these preferences?

- ☐ $u = 2x + 3y$
- ☐ $u = 3x + 2y$
- ☐ $u = \min\{2x, 3y\}$
- ☐ $u = x + y$
- ☐ $u = xy$

(c) One hour of running burns 600 calories and one hour of cycling burns 400 calories. The consumer cares only about the total number of calories burned. Let x denote the hours of running and y the hours of cycling. Consider these utility functions.

I. $u = 600x + 400y$
II. $u = 3x + 2y$
III. $u = \min\{600x, 400y\}$

Which of the utility functions above represent these preferences?

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

Part B · More preferences in words

Each story below describes a consumer's preferences in words.

(a) A hot dog requires one sausage and one bun. A sausage without a bun yields no utility, and likewise a bun without a sausage. Let x denote the number of sausages and y the number of buns. The consumer cares only about the number of hot dogs. Which utility function represents these preferences?

- ☐ $u = \min\{3x, y\}$
- ☐ $u = \min\{x, 3y\}$
- ☐ $u = \min\{x, y\}$
- ☐ $u = x + y$
- ☐ $u = xy$

(b) Each sandwich requires 1 slice of bread and exactly 3 slices of cheese. Let x denote the number of slices of bread and y the number of slices of cheese. The consumer cares only about the number of sandwiches. Which utility function represents these preferences?

- ☐ $u = x + 3y$
- ☐ $u = \min\{\frac{x}{3}, y\}$
- ☐ $u = \min\{x, 3y\}$
- ☐ $u = \min\{x, \frac{y}{3}\}$
- ☐ $u = x + \frac{y}{3}$

(c) A cake requires 3 eggs and 5 spoons of sugar, always in that ratio. Let x denote the number of eggs and y the number of spoons of sugar. The consumer cares only about the number of cakes. Consider these utility functions.

I. $u = \min\{\frac{x}{3}, \frac{y}{5}\}$
II. $u = \min\{5x, 3y\}$
III. $u = \min\{3x, 5y\}$

Which of the utility functions above represent these preferences?

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

Part C · Budget lines in words

Each story below describes a consumer's budget in words. The consumer spends the entire income.

(a) A consumer has €24. The price of a coffee is €2 and the price of a sandwich is €3. Let x denote the number of coffees and y the number of sandwiches. Which equation describes the consumer's budget line?

- ☐ $2x + 3y = 12$
- ☐ $2x + 3y = 24$
- ☐ $2x + 3y = 5$
- ☐ $3x + 2y = 24$
- ☐ $x + y = 24$

(b) A consumer has €30. The first 5 gym visits cost €4 each, and each further visit costs €2. All other goods cost €1 per unit. Let x denote the number of visits and y the quantity of all other goods. Which equations describe the consumer's budget line?

- ☐ $2x + y = 30$ for $x \leq 5$, and $4x + y = 30$ for $x \geq 5$
- ☐ $4x + y = 30$ for $x \leq 4$, and $2x + y = 20$ for $x \geq 4$
- ☐ $4x + y = 30$ for $x \leq 5$, and $2x + y = 10$ for $x \geq 5$
- ☐ $4x + y = 30$ for $x \leq 5$, and $2x + y = 20$ for $x \geq 5$
- ☐ $4x + y = 30$ for $x \leq 5$, and $2x + y = 30$ for $x \geq 5$

(c) A consumer has €20. The first 3 bus rides are free, and each further ride costs €2. The consumer can take at most 10 rides. All other goods cost €1 per unit. Let x denote the number of rides and y the quantity of all other goods. Which equations describe the consumer's budget line?

- ☐ $2x + y = 20$ for $x \leq 10$
- ☐ $y = 20$ for $x \leq 3$, and $2x + y = 20$ for $3 \leq x \leq 10$
- ☐ $y = 20$ for $x \leq 3$, and $2x + y = 23$ for $3 \leq x \leq 10$
- ☐ $y = 20$ for $x \leq 3$, and $2x + y = 26$ for $3 \leq x \leq 10$
- ☐ $y = 20$ for $x \leq 3$, and $2x + y = 26$ for $x \geq 3$, with no limit on the number of rides